

Course Code	Name of the Course			
216U03C301	Mathematics For Communication Engineering			
Teaching Scheme	TH	P	TUT	Total
	03	0	01	04
Credits Assigned	03	0	01	04
Evaluation Scheme	Marks			
	LAB/TUT CA	CA (TH)		ESE
		IA	ISE	
	25	20	30	50
				125

Course pre-requisites:

- Applied Mathematics-I, Applied Mathematics-II

Course Objectives:

The objective of this course is to impart knowledge of Transform & Vector Calculus, concepts of Fourier Series and Bessel functions and thereby aid the students to apply the fundamental of mathematics necessary to formulate, solve and analyze Electronics and telecommunication engineering problems in the areas related to Electrical Network analysis, Communication Systems, Control systems, Signals and Systems & Electromagnetic Field Theory.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

- CO1.** Find Laplace Transform, Inverse Laplace Transform of function & Apply Laplace Transform to solve Differential Equations.
- CO2.** Find Fourier Series representation for a periodic function.
- CO3.** Solve problems on recurrence relation of Bessel function.
- CO4.** Solve examples using operators grad, div & curl.
- CO5.** Solve examples to evaluate integrals using Green's Theorem, Stokes' Theorem & Divergence theorem.

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs.	CO
1	Laplace Transform		12	CO1
	1.1	Definition of Laplace Transform, Laplace Transform of $\sin(at)$, $\cos(at)$, $\sinh(at)$, $\cosh(at)$, $\text{erf}(t)$, Heavi-side unit step, dirac-delta function, Laplace Transform of periodic function		
	1.2	Properties of Laplace Transform @ Linearity, first shifting theorem, second shifting theorem, multiplication by t , division by t , Laplace Transform of derivatives and integrals, change of scale.		
	1.3	Inverse Laplace Transform: Partial fraction method, convolution theorem @		
	1.4	Applications of Laplace Transform: Solution of ordinary differential equations with constant coefficients.		
2	Fourier Series		10	CO2
	2.1	Introduction: Definition, Dirichlet’s conditions, Euler’s formulae		
	2.2	Fourier Series of Functions: Exponential, trigonometric functions, even and odd functions, half range sine and Cosine series		
	2.3	Complex form of Fourier series		
3	Bessel Functions		06	CO3
	3.1	Bessel’s Differential Equation and Bessel function		
	3.2	Recurrence relation, Properties of Bessel function @.		
4	Vector Differentiation		08	CO4
	4.1	Scalar and vector product of three and four vectors and their properties.		
	4.2	Gradient of scalar point function, divergence and curl of vector point function.		
	4.3	Solenoidal and irrotational vector fields.		
#Self-Learning: Co-ordinate Systems and its conversion.				
5	Vector Integration		09	CO5
	5.1	Vector Integral: Line integral, Properties of line integral, Surface integral, Volume integrals.		
	5.2	Green’s theorem in a plane @		
	5.3	Gauss divergence theorem, Stokes theorem @		
Total			45	

@ Proofs of properties/ theorems are not expected

Students should prepare self-learning topics on their own. Self – learning topics will enable students to gain extended knowledge of the topic. Assessment of these topics may be included in IA.

Reference Books*

Sr No	Name/s of Author/s	Title of Book	Publisher	Edition/ Year
1	B. S. Grewal	<i>Higher Engineering Mathematics</i>	Khanna Publications, India	43rd Edition, 2014
2	Erwin Kreyszig	<i>Advanced Engineering Mathematics</i>	Wiley Eastern Limited, India	10th Edition 2015
3	P.N.Wartikar & J. N. Wartikar,	<i>A Text Book of Applied Mathematics, Vol I & Vol II</i>	Vidyarthi Griha Prakashan,	6th Edition 2012
4	B.V.Ramana	<i>Higher Engineering Mathematics</i>	Tata McGrawHill Publishing Company Limited, India	1st Edition 2007
5	N.P.Bali, Dr.Manish Goyal.	<i>A Textbook of Engineering Mathematics.</i>	Laxmi Publication, India	9th Edition 2016

Tutorial: Tutorial will consist of at least 10 Tutorials covering entire syllabus out of which 3 to 4 tutorials can be Telecommunication application based. Use of some Programming Language/software is also recommended to conduct Tutorials. Students will be graded based on continuous assessment of their term work